

Competitive Programming Junior Sheet

A structured roadmap of problems, video lessons and solutions for newcomers to competitive programming

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Version: V5.6 (as recorded in the source sheet)

Scope: ~800 problems for newcomers to problem solving, covering all topics needed up to Codeforces Div2-D

For whom: A “junior” is anyone who does not yet master solving Codeforces Div2-D

Key links

[Ask.fm — Q&A with the author](#)

[Author site / more contacts](#)

[Video explaining this sheet \(Arabic, a bit outdated\)](#)

[Prerequisite: C++ Programming playlist](#)

[Why C++ is recommended](#)

[Assiut University beginner problem set](#)

[About the community & programming competitions](#)

[Solve To Be channel \(Java\)](#)

[Code El Masry channel \(Java\)](#)

External resources

[Awesome Competitive Programming \(curated link list\)](#)

[Ahmed Elsaghir training curriculum](#)

[A2OJ Ladders \(alternative to this sheet\)](#)

[Programming playlist — Ahmed M. Sayed \(Arabic\)](#)

[Programming playlist — Mohamed Desouky \(Arabic\)](#)

[More resources playlist](#)

How to use this guide: each section below corresponds to one training sheet (Div2-A through Div2-D). Rows marked **WATCH** are lesson videos to watch before attempting the problems that follow them. Each problem row links directly to the problem statement, and, where available, a companion video solution, editorial, or reference solution.

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Sheet A — Codeforces Div2-A (Absolute Beginner) (216 entries)

Sheet B — Codeforces Div2-B (186 entries)

Sheet C1 — Codeforces Div2-C (Part 1) (213 entries)

Sheet C2 — Codeforces Div2-C (Part 2) (185 entries)

Sheet D1 — Codeforces Div2-D (Part 1) (140 entries)

Sheet D2 — Codeforces Div2-D (Part 2) (131 entries)

Sheet D3 — Codeforces Div2-D (Part 3) (124 entries)

Frequently Asked Questions

Frequently Asked Questions

What programming background do I need?

Only programming skills (e.g. a first programming course level). No OOP or data-structures background required, though learning STL (or Java/C# Collections) helps a lot. Implementing 2-3 small projects first is advised.

More on prerequisites

<https://ask.fm/mostafasaad87/answers/144907000290>

What should I study after finishing sheet CF-C2?

Finish up to the CF-C2 sheet, then study from the book "Cracking the Coding Interview: 150 Programming Questions and Solutions".

<https://www.youtube.com/watch?v=39vqarATPyM>

How were problems rated / ordered?

Many sheet versions existed, each improving on the last. Codeforces problems were rated using a community rating tool.

<http://codeforces.com/blog/entry/46304?mobile=true>

Where can I find more on the community / competitions?

<https://www.youtube.com/playlist?list=PLPt2dINI2MlNcU070HIAO8JWYBcafuyG>

More Q&A from the author

<https://ask.fm/mostafasaad87/answers/145333554402>

Should I delay competitive programming to focus on graduation?

The author's advice: graduate on time, and use free time and vacations to do more problem solving.

<https://ask.fm/mostafasaad87/answers/145151822818>

Sheet A — Codeforces Div2-A (Absolute Beginner)

Problem	Problem Link	Video / Solution
Sample Name1	—	Solution in mind is $O(n^2)$: for each pair of points, get the equation of the straight line linking between them, and add 2 to its count. Print the count of the line having the max
Sample Name2	—	I had to check the editorial My Performance Notes: This is so bad performance. He thought for little time and continued thinking while coding. As a result, much debug time too. RU
Sample Name3	—	Please always write and study your timings. My Performance Notes: This is so good performance. 1st submission. Thinking is the higher. Code/Debug is so low. By time, thinking colu
Sample Name4	—	Other Status values: AC, WA, CS, TLE, MLE, RTE, ... These values and comments are just examples. Just remove/ignore them.
Sample Name5	—	Want c++ solution for UVA 408? Google with: UVA 408 filetype:c++

■ **WATCH:** [Watch - Approaching Problem Statement](#)

■ **WATCH:** [Watch - Thinking - On papers Not on PC](#)

Problem	Problem Link	Video / Solution
Vanya and Fence	CF677-D2-A	C++ Solution Example
Anton and Danik	CF734-D2-A	This is from Round 379. Here is the editorial
Bear and Big Brother	CF791-D2-A	Video Solution - Eng Youssef El Ghareeb
Team	CF231-D2-A	Video Solution - Eng Youssef Ali
Beautiful Matrix	CF263-D2-A	Video Solution - Eng Samed Hajaila
Gravity Flip	CF405-D2-A	Video Solution - Eng John Gamal
Petya and Strings	CF112-D2-A	Video Solution - Solver to be (Java)
Boy or Girl	CF236-D2-A	Video Solution - Solver to be (Java)
Word	CF59-D2-A	Video Solution - Solver to be (Java)
Word Capitalization	CF281-D2-A	Video Solution - Solver to be (Java)
Magnets	CF344-D2-A	Video Solution - Solver to be (Java)
Sereja and Dima	CF381-D2-A	Video Solution - Solver to be (Java)
Stones on the Table	CF266-D2-A	Video Solution - Eng Ahmead Raafat (Python)
Police Recruits	CF427-D2-A	Video Solution - Eng Ahmead Raafat (Python)
Black Square	CF431-D2-A	—
Night at the Museum	CF731-D2-A	Video Solution - Eng Yahia Ashraf
Games	CF268-D2-A	Video Solution - Eng Yahia Ashraf
Buy a Shovel	CF732-D2-A	Video Solution - Eng Yahia Ashraf

Problem	Problem Link	Video / Solution
Is your horseshoe on the other hoof?	CF228-D2-A	Video Solution - Eng Ahmead Raafat (Python)
Colorful Stones (Simplified Edition)	CF265-D2-A	Video Solution - Eng Ahmead Raafat (Python)

■ **WATCH:** [Watch - Measuring Algorithms Performance - 1](#)

■ **WATCH:** [Watch - Elementary Math - Introduction](#)

Problem	Problem Link	Video / Solution
Die Roll	CF9-D2-A	Video Solution - Eng Muntaser Abukadeja
Shaass and Oskols	CF294-D2-A	Video Solution - Eng Mostafa Saad
Juicer	CF709-D2-A	Video Solution - Solver to be (Java)
Carrot Cakes	CF799-D2-A	Video Solution - Solver to be (Java)
Anton and Letters	CF443-D2-A	Video Solution - Solver to be (Java)
Way Too Long Words	CF71-D2-A	Video Solution - Solver to be (Java)
Free Ice Cream	CF686-D2-A	Video Solution - Solver to be (Java)
Helpful Maths	CF339-D2-A	Video Solution - Solver to be (Java)
Team Olympiad	CF490-D2-A	Video Solution - Eng Muntaser Abukadeja
New Password	CF770-D2-A	Video Solution - Solver to be (Java)

■ **WATCH:** [Watch - Number Theory - Modular Arithmetic](#)

■ **WATCH:** [Watch - Combinatorics - Counting Principles](#)

Problem	Problem Link	Video / Solution
Light, more light	UVA 10110	Video Solution - Eng Amr Saud
Product	UVA 10106	Video Solution - Eng Youssef El Ghareeb. Don't solve using big integer
Uniform Generator	UVA 408	Video Solution - Eng Yahia Ashraf
Black and white painting	UVA 11231	Video Solution - Eng Amr Saud
Presents	CF136-D2-A	—
Lineland Mail	CF567-D2-A	Video Solution - Eng Ahmed Rafaat (Python)
Mahmoud and Longest Uncommon Subsequence	CF766-D2-A	Video Solution - Solver to be (Java)
Snacktower	CF767-D2-A	Video Solution - Solver to be (Java)
Oath of the Night's Watch	CF768-D2-A	Video Solution - Solver to be (Java)
Next Round	CF158-D12-A	Video Solution - Solver to be (Java)
Bit++	CF282-D2-A	Video Solution - Solver to be (Java)
Young Physicist	CF69-D2-A	Video Solution - Solver to be (Java)
Pangram	CF520-D2-A	Video Solution - Solver to be (Java)
Twins	CF160-D2-A	Video Solution - Solver to be (Java)
Keyboard	CF474-D2-A	Video Solution - Solver to be (Java)

■ **WATCH:** [Watch - Graph Theory - Intro](#)

■ **WATCH:** [Watch - Graph Theory - DFS](#)

Problem	Problem Link	Video / Solution
The Seasonal War	UVA 352	Video Solution - Eng Mohamed Nasser
Marcus	UVA 10452	Video Solution - Eng Ayman Salah
Battleships	UVA 11953	Video Solution - Eng Aya Elymany
Forming Teams	CF216-D2-B	Video Solution - Eng Mostafa Saad
Hierarchy	SPOJ MAKETREE	—
Ordering Tasks	UVA 10305	—
Even Odds	CF318-D2-A	Video Solution - Eng Muntaser Abukadeja
I Wanna Be the Guy	CF469-D2-A	Video Solution - Solver to be (Java)
Is it rated?	CF807-D2-A	Video Solution - Solver to be (Java)
Olesya and Rodion	CF584-D2-A	Video Solution - Solver to be (Java)
String Task	CF118-D2-A	Video Solution - Solver to be (Java)
Translation	CF41-D2-A	Video Solution - Solver to be (Java)
Case of the Zeros and Ones	CF556-D2-A	Video Solution - Solver to be (Java)
Football	CF43-D2-A	Video Solution - Solver to be (Java)
Brain's Photos	CF707-D2-A	Video Solution - Solver to be (Java)
Dubstep	CF208-D2-A	Video Solution - Solver to be (Java)

■ **WATCH: [Watch - Computational Geometry - Intro](#)**

■ **WATCH: [Watch - Computational Geometry - Point and Vector](#)**

Problem	Problem Link	Video / Solution
Wasted Time	CF127-D2-A	—
Points in Figures: Rectangles	UVA 476	—
Overlapping Rectangles	UVA 460	Video Solution - Eng Muntaser Abukadeja
Fancy Fence	CF270-D2-A	—
Pouring Rain	CF667-D2-A	—
Fourth Point !!	UVA 10242	Video Solution - Eng Magdy Hasan
Valera and X	CF404-D2-A	Video Solution - Solver to be (Java)
Arpa's hard exam and Mehrdad's naive cheat	CF742-D2-A	Video Solution - Solver to be (Java)
Calculating Function	CF486-D2-A	Video Solution - Solver to be (Java)
Theatre Square	CF1-D12-A	Video Solution - Solver to be (Java)
Anton and Polyhedrons	CF785-D2-A	Video Solution - Solver to be (Java)
Panoramix's Prediction	CF80-D2-A	Video Solution - Solver to be (Java)
Counterexample	CF483-D2-A	Video Solution - Solver to be (Java)
Good Number	CF365-D2-A	Video Solution - Eng Muntaser Abukadeja
Dice Tower	CF225-D2-A	Video Solution - Eng Muntaser Abukadeja
Alyona and Numbers	CF682-D2-A	Video Solution - Eng John Gamal
Mountain Scenery	CF218-D2-A	Video Solution - Eng John Gamal
Help Vasilisa the Wise 2	CF143-D2-A	Video Solution - Eng John Gamal
Chewba■ca and Number	CF514-D2-A	Video Solution - Eng Muntaser Abukadeja
Ksenia and Pan Scales	CF382-D2-A	Video Solution - Eng Samed Hajaila
Launch of Collider	CF699-D2-A	Video Solution - Eng Samed Hajaila

Problem	Problem Link	Video / Solution
Polo the Penguin and Segments	CF289-D2-A	Video Solution - Eng Mostafa Saad
IQ Test	CF287-D2-A	Video Solution - Eng Mostafa Saad
Yaroslav and Permutations	CF296-D2-A	Video Solution - Eng Mostafa Saad
Laptops	CF456-D2-A	Video Solution - Solver to be (Java)
Snow Footprints	CF298-D2-A	Video Solution - Eng Mostafa Saad
Raising Bacteria	CF579-D2-A	Video Solution - Eng Ahmed Rafaat (Python)

■ **WATCH: [Watch - Search Techniques - Binary Search](#)**

Problem	Problem Link	Video / Solution
The Playboy Chimp	UVA 10611	Video Solution - Eng Ayman Salah
Pipeline	CF287-D2-B	Video Solution - Eng Mostafa Saad
Aggressive cows	SPOJ AGGRCOW	Video Solution - Eng Youssef El Ghareeb
Left-handers, Right-handers and Ambidexters	CF950-D2-A	Video Solution - Eng Hossam Yehia
George and Accommodation	CF467-D2-A	—
Vasya the Hipster	CF581-D2-A	—
Fox And Snake	CF510-D2-A	—
The New Year: Meeting Friends	CF723-D2-A	—
Elephant	CF617-D2-A	—
Greg's Workout	CF255-D2-A	—
Ultra-Fast Mathematician	CF61-D2-A	—
Little Pony and Crystal Mine	CF454-D2-A	—
One-dimensional Japanese Crossword	CF721-D2-A	—
Soldier and Bananas	CF546-D2-A	—
Bus to Udayland	CF711-D2-A	—
Cookies	CF129-D2-A	—
Second Order Statistics	CF22-D2-A	—
Nearly Lucky Number	CF110-D2-A	—
Playing with Dice	CF378-D2-A	—
A Good Contest	CF681-D2-A	—
Beautiful Year	CF271-D2-A	—
Far Relative's Birthday Cake	CF629-D2-A	—
Mashmokh and Lights	CF415-D2-A	—
Triangular numbers	CF47-D2-A	—
Roma and Lucky Numbers	CF262-D2-A	—
Toy Army	CF84-D2-A	—
Levko and Table	CF361-D2-A	—
Cards	CF701-D2-A	—
Wizards' Duel	CF591-D2-A	—
Combination Lock	CF540-D2-A	—
Summer Camp	CF672-D2-A	—

Problem	Problem Link	Video / Solution
Soft Drinking	CF151-D2-A	—
Coder	CF384-D2-A	—
GukiZ and Contest	CF551-D2-A	—
Circle Line	CF278-D2-A	—
Patrick and Shopping	CF599-D2-A	—
Choosing Teams	CF432-D2-A	—
Vanya and Cubes	CF492-D2-A	—
Insomnia cure	CF148-D2-A	—
Cakeminator	CF330-D2-A	—
Flag	CF16-D2-A	—
Cupboards	CF248-D2-A	—
Soroban	CF363-D2-A	—
Amusing Joke	CF141-D2-A	—
Lights Out	CF275-D2-A	—
Lunch Rush	CF276-D2-A	—
Duff and Meat	CF588-D2-A	—
Vanya and Cards	CF401-D2-A	—
Squats	CF424-D2-A	—
Arrival of the General	CF144-D2-A	—
Sinking Ship	CF63-D2-A	—
LLPS	CF202-D2-A	—
Candy Bags	CF334-D2-A	—
Game With Sticks	CF451-D2-A	—
Vasya and Socks	CF460-D2-A	—
Dima and Friends	CF272-D2-A	—
Nicholas and Permutation	CF676-D2-A	—
Toy Cars	CF545-D2-A	—
DZY Loves Hash	CF447-D2-A	—
HQ9+	CF133-D2-A	—
Holidays	CF670-D2-A	—
Dividing Orange	CF244-D2-A	—
Haiku	CF78-D2-A	—
System of Equations	CF214-D2-A	—
IQ test	CF25-D2-A	—
Contest	CF501-D2-A	—
Restoring Password	CF94-D2-A	—
Valera and Plates	CF369-D2-A	—
Minimum Difficulty	CF496-D2-A	—
Little Elephant and Function	CF221-D2-A	—
Collecting Beats is Fun	CF373-D2-A	—

Problem	Problem Link	Video / Solution
Letter	CF14-D2-A	—
Kefa and First Steps	CF580-D2-A	—
Ilya and Bank Account	CF313-D2-A	—
Uncowed Forces	CF604-D2-A	—
Reconnaissance 2	CF34-D2-A	—
Lucky Ticket	CF146-D2-A	—
Chat room	CF58-D2-A	—
George and Sleep	CF387-D2-A	—
Ostap and Grasshopper	CF735-D2-A	—
The number of positions	CF124-D2-A	—
Table	CF359-D2-A	—
Tavas and Nafas	CF535-D2-A	—
Watermelon	CF4-D2-A	—
Let's Watch Football	CF195-D2-A	—
Initial Bet	CF478-D2-A	—
Saitama Destroys Hotel	CF608-D2-A	—
Queue on Bus Stop	CF435-D2-A	—
Bicycle Chain	CF215-D2-A	—
Little Elephant and Rozdil	CF205-D2-A	—
Amr and Music	CF507-D2-A	—
Marks	CF152-D2-A	—
Postcards and photos	CF137-D2-A	—
Business trip	CF149-D2-A	—
Drazil and Date	CF515-D2-A	—
Multiplication Table	CF577-D2-A	—
Exam	CF534-D2-A	—
Alena's Schedule	CF586-D2-A	—
Interview	CF631-D2-A	—
Lucky Division	CF122-D2-A	—
Appleman and Easy Task	CF462-D2-A	—
Vasya and Digital Root	CF355-D2-A	—
Parallelepiped	CF224-D2-A	—
Group of Students	CF357-D2-A	—
Joysticks	CF651-D2-A	—
Array	CF300-D2-A	—
Round House	CF659-D2-A	—
Lala Land and Apple Trees	CF558-D2-A	—
Autocomplete	CF53-D2-A	—
Digital Counter	CF495-D2-A	—
Vitaliy and Pie	CF525-D2-A	—

Problem	Problem Link	Video / Solution
Life Without Zeros	CF75-D2-A	—

Sheet B — Codeforces Div2-B

■ **WATCH:** [Watch - Thinking - Problem Simplification](#)

■ **WATCH:** [Watch - Thinking - Brainstorm - Rank - Approach](#)

■ **WATCH:** [Watch - Combinatorics - Permutations and Combinations - 1](#)

■ **WATCH:** [Watch - Combinatorics - Permutations and Combinations - 2](#)

Problem	Problem Link	Video / Solution
Decoding	CF746-D2-B	Video Solution - Solver to be (Java)
Petya and Countryside	CF66-D2-B	Video Solution - Eng Muntaser Abukadeja
Bear and Finding Criminals	CF680-D2-B	Video Solution - Eng Muntaser Abukadeja
Burglar and Matches	CF16-D2-B	Video Solution - Eng Muntaser Abukadeja
Caixa and Pylons	CF463-D2-B	Video Solution - Eng Muntaser Abukadeja
Sum of Digits	CF102-D2-B	Video Solution - Eng Muntaser Abukadeja
Coins	CF47-D2-B	Video Solution - Eng Samed Hajajla
Vanya and Lanterns	CF492-D2-B	Video Solution - Solver to be (Java)
Effective Approach	CF227-D2-B	Video Solution - Eng Abanob Ashraf
Easter Eggs	CF78-D2-B	Video Solution - Eng Abanob Ashraf

■ **WATCH:** [Watch - Training-Secrets of Success](#)

■ **WATCH:** [Watch - Number Theory - Fib, GCD, LCM, Pow](#)

Problem	Problem Link	Video / Solution
Big Mod	UVA 374	—
Euclid's Game	UVA 10368	Video Solution - Eng Moaz Rashad
Combinations	UVA 369	—
Pi	UVA 412	Video Solution - Eng Mohamed Adel
Adding Reversed Numbers	UVA 713	Don't use big integer class. Write simple array computations
President's Office	CF6-D2-B	Video Solution - Eng Muntaser Abukadeja
Fence	CF363-D2-B	Video Solution - Eng Muntaser Abukadeja
Lovely Palindromes	CF688-D2-B	Video Solution - Solver to be (Java)
Sort the Array	CF451-D2-B	Video Solution - Solver to be (Java)
Devu, the Dumb Guy	CF439-D2-B	Video Solution - Solver to be (Java)
Summer sell-off	CF810-D2-B	Video Solution - Solver to be (Java)
Colorful Field	CF79-D12-B	Video Solution - Solver to be (Java)
Keyboard	CF88-D2-B	Video Solution - Eng Muntaser Abukadeja
Mahmoud and a Triangle	CF766-D2-B	Video Solution - Solver to be (Java)
Find The Bone	CF796-D2-B	Video Solution - Solver to be (Java)

■ **WATCH:** [Watch - Graph Theory - BFS](#)

Problem	Problem Link	Video / Solution
Tic-Tac-Toe (I)	SPOJ TOE1	Video Solution - Eng Ayman Salah
Tic-Tac-Toe (II)	SPOJ TOE2	Video Solution - Eng Essam AlNaggar

Problem	Problem Link	Video / Solution
Knight Moves	UVA 439	Video Solution - Eng Magdy Hasan
King's Path	CF242-D2-C	Video Solution - Eng Mostafa Saad
Countdown	SPOJ CDOWN	Video Solution - Solver to be (Java)
Students and Shoelaces	CF129-D2-B	Video Solution - Eng Abanob Ashraf
Dreamoon and WiFi	CF476-D2-B	Video Solution - Eng Mohamed Adel
Chat Online	CF469-D2-B	Video Solution - Eng Mohamed Adel
Olympic Medal	CF215-D2-B	Video Solution - Eng Ahmed Salah
Filya and Homework	CF714-D2-B	Video Solution - Eng Muntaser Abukadeja
Inna and New Matrix of Candies	CF400-D2-B	Video Solution - Eng Mohamed Salah
Steps	CF152-D2-B	Video Solution - Eng Muntaser Abukadeja
Growing Mushrooms	CF186-D2-B	Video Solution - Eng Mohamed Salah
Regular Bracket Sequence	CF26-D12-B	Video Solution - Solver to be (Java)
Escape	CF148-D2-B	—

■ **WATCH: [Watch - Intro to DP - 1](#)**

■ **WATCH: [Watch - Intro to DP - 2](#)**

Problem	Problem Link	Video / Solution
Vacations	CF699-D2-C	—
Woodcutters	CF545-D2-C	—
Divisibility by Eight	CF550-D2-C	—
Barcode	CF225-D2-C	Video Solution - Eng Mostafa Saad
Continents	UVA 11094	Video Solution - Eng Ayman Salah
Brownie Points	UVA 10865	Video Solution - Eng Magdy Hasan
Roma and Changing Signs	CF262-D2-B	Video Solution - Eng Mohamed Salah
Bear and Strings	CF385-D2-B	—
I.O.U.	CF376-D2-B	Video Solution - Eng Abanob Ashraf
Jeff and Periods	CF352-D2-B	—
Meeting	CF144-D2-B	—
Chocolate	CF617-D2-B	—
Easy Number Challenge	CF236-D2-B	—
Han Solo and Lazer Gun	CF514-D2-B	—
Physics Practical	CF253-D2-B	—
Two Buttons	CF520-D2-B	Video Solution - Solver to be (Java)

■ **WATCH: [Watch - Computational Geometry - Complex Number and 2D Point](#)**

■ **WATCH: [Watch - Computational Geometry - Lines and Distances](#)**

Problem	Problem Link	Video / Solution
Intersecting Lines	UVA 378	—
The Stern-Brocot Number	UVA 10077	—
Garland	UVA 1555	—
Mr. Kitayuta's Colorful Graph	CF505-D2-B	Video Solution - Eng Muntaser Abukadeja

Problem	Problem Link	Video / Solution
Fox Dividing Cheese	CF371-D2-B	Video Solution - Eng Abanob Ashraf
DZY Loves Chemistry	CF445-D2-B	—
Kolya and Tanya	CF584-D2-B	—
Suffix Structures	CF448-D2-B	—
Complete the Word	CF716-D2-B	—
Sea and Islands	CF544-D2-B	—
Hopscotch	CF141-D2-B	—
Valera and Contest	CF369-D2-B	—
Bear and Friendship Condition	CF791-D2-B	Video Solution - SolverToBe (Java)
Preparing Olympiad	CF550-D2-B	Video Solution - SolverToBe (Java)

■ **WATCH: Watch - Focused and Diffused Thinking**

■ **WATCH: Watch - Graph Theory - MST - Kruskal**

Problem	Problem Link	Video / Solution
ACM contest and Blackout	UVA 10600	Video Solution - Eng Moaz Rashad
Highways	UVA 10147	Video Solution - Eng Mahmoud Adel
Virtual Friends	UVA 11503	Video Solution - Eng Moaz Rashad
Arctic Network	UVA 10369	—
Trees on the level	UVA 122	Video Solution - SolverToBe (Java)
Rankings	UVA 12263	Editorial to read
Mashmokh and Tokens	CF415-D2-B	—
Approximating a Constant Range	CF602-D2-B	—
Gena's Code	CF614-D2-B	—
OR in Matrix	CF486-D2-B	—
Fox And Two Dots	CF510-D2-B	Video Solution - Eng Mohamed Adel
Routine Problem	CF337-D2-B	Video Solution - Eng Mohamed Adel
Vasya and Wrestling	CF493-D2-B	—
Hamming Distance Sum	CF608-D2-B	—
Wet Shark and Bishops	CF621-D2-B	—
Kefa and Company	CF580-D2-B	Video Solution - SolverToBe (Java)
Tavas and SaDDas	CF535-D2-B	Video Solution - Eng Abanob Ashraf

■ **WATCH: Watch - Intro to Greedy**

Problem	Problem Link	Video / Solution
Painting Eggs	CF282-D2-B	—
Pasha Maximizes	CF435-D2-B	Video Solution - Eng Hossam Yehia
Little Girl and Game	CF276-D2-B	Video Solution - Eng Hossam Yehia
Pasha and String	CF525-D2-B	Video Solution - Eng Hossam Yehia
Booking System	CF416-D2-C	—
Vanya and Exams	CF492-D2-C	—
Chocolate	SPOJ CHOCOLA	—

Problem	Problem Link	Video / Solution
The Skyline Problem	UVA 105	—
Hanoi Tower Troubles !	UVA 10276	Video Solution - Eng Mahmoud Adel
Maze Exploration	UVA 784	Video Solution - Eng Mahmoud Adel
IP-TV	—	—
Inbox (100500)	CF465-D2-B	—
Different is Good	CF672-D2-B	—
Permutation	CF137-D2-B	—
Little Elephant and Magic Square	CF259-D2-B	—
Airport	CF218-D2-B	—
Cormen --- The Best Friend Of a Man	CF732-D2-B	—
Prison Transfer	CF427-D2-B	—
A and B and Compilation Errors	CF519-D2-B	—
Letter	CF43-D2-B	—
Game of Robots	CF670-D2-B	—
African Crossword	CF90-D2-B	—
Cows and Poker Game	CF284-D2-B	—
Find Marble	CF285-D2-B	—
Interesting drink	CF706-D2-B	—
Megacity	CF424-D2-B	—
Beautiful Paintings	CF651-D2-B	—
Ilya and Queries	CF313-D2-B	—
Code Parsing	CF255-D2-B	—
Hungry Sequence	CF327-D2-B	—
Chloe and the sequence	CF743-D2-B	—
Luxurious Houses	CF581-D2-B	—
Settlers' Training	CF63-D2-B	—
Far Relative's Problem	CF629-D2-B	—
Wilbur and Array	CF596-D2-B	—
Text Document Analysis	CF723-D2-B	—
Shower Line	CF431-D2-B	—
Misha and Changing Handles	CF501-D2-B	—
Coat of Anticubism	CF667-D2-B	—
Ternary Logic	CF136-D2-B	—
Counting Rhombi	CF189-D2-B	—
Pashmak and Flowers	CF459-D2-B	—
The Monster and the Squirrel	CF592-D2-B	—
The Fibonacci Segment	CF365-D2-B	—
Spider Man	CF705-D2-B	—
Little Robber Girl's Zoo	CF686-D2-B	—
Unary	CF133-D2-B	—

Problem	Problem Link	Video / Solution
Canvas Frames	CF127-D2-B	—
Ohana Cleans Up	CF554-D2-B	—
Garland	CF408-D2-B	—
Petya and Staircases	CF362-D2-B	—
Equidistant String	CF545-D2-B	—
Vanya and Food Processor	CF677-D2-B	—
Calendar	CF304-D2-B	—
Amr and Pins	CF507-D2-B	—
Polo the Penguin and Matrix	CF289-D2-B	—
George and Round	CF387-D2-B	—
Alyona and flowers	CF740-D2-B	—
Urbanization	CF735-D2-B	—
Testing Pants for Sadness	CF104-D2-B	—
Cells Not Under Attack	CF701-D2-B	—
Vanya and Books	CF552-D2-B	—
Worms	CF474-D2-B	—
Kuriyama Mirai's Stones	CF433-D2-B	—
Fortune Telling	CF59-D2-B	—
Walking in the Rain	CF192-D2-B	—
Dima and To-do List	CF366-D2-B	—
Sail	CF298-D2-B	—
Fox and Cross	CF389-D2-B	—
Rebranding	CF591-D2-B	—
Increase and Decrease	CF246-D2-B	—
Alyona and Mex	CF682-D2-B	—
Coins	CF58-D2-B	—
Berland National Library	CF567-D2-B	—
Art Union	CF416-D2-B	—
BerSU Ball	CF489-D2-B	—
Random Teams	CF478-D2-B	—
Friends	CF94-D2-B	—
War of the Corporations	CF625-D2-B	—
Road Construction	CF330-D2-B	—
Binary Number	CF92-D2-B	—
Before an Exam	CF4-D2-B	—
Running Student	CF9-D2-B	—
Anton and currency you all know	CF508-D2-B	—
Phone Numbers	CF151-D2-B	—

Sheet C1 — Codeforces Div2-C (Part 1)

Problem	Problem Link	Video / Solution
Drazil and Factorial	CF515-D2-C	Video Solution - Eng Mostafa Saad
Sagheer and Nubian Market	CF812-D2-C	Video Solution - Solver to be (Java)
Closing ceremony	CF720-D1-A	Video Solution - Solver to be (Java)
Petya and Java	CF66-D2-A	Video Solution - Solver to be (Java)
Vanya and Fence	CF67-D2-A	Video Solution - Solver to be (Java)
Lucky Permutation Triple	CF304-D2-C	—
Fire Again	CF35-D2-C	—
Soldier and Cards	CF546-D2-C	—
Watchmen	CF651-D2-C	—
Ilya and Matrix	CF313-D2-C	—
The Child and Set	CF437-D2-B	—
Tanya and Postcard	CF518-D2-B	—
Mike and Fun	CF548-D2-B	—

■ **WATCH:** [Watch - Thinking - Concretely - Symbolically - Pictorially](#)

■ **WATCH:** [Watch - Thinking - Problem Constraints](#)

■ **WATCH:** [Watch - Number Theory - Primes](#)

Problem	Problem Link	Video / Solution
Fox Dividing Cheese	CF371-D2-B	—
Duff in Love	CF588-D2-B	—
Twin Primes	UVA 10394	—
Summation of Four	UVA 10168	Video Solution - Eng Moaz Rashad
The Lottery	UVA 10325	—
Mint	UVA 10717	—
BITMAP - Bitmap	SPOJ BITMAP	—
Key Task	SPOJ CERC07K	—
—	—	Revise Probability
Sonya and Queries	CF714-D2-C	—
Median	CF166-D2-C	—
Learning Languages	CF278-D2-C	—
Petya and Inequations	CF112-D2-C	—
Beautiful Sets of Points	CF268-D2-C	—
History	CF137-D2-C	Video Solution - Solver to be (Java)
Tanya and Toys	CF659-D2-C	Video Solution - Solver to be
Tram	CF746-D2-C	Video Solution - Solver to be (Java)
The World is a Theatre	CF131-D2-C	Video Solution - Eng Youssef Ali
Trains	CF88-D2-C	Video Solution - Solver to be (Java)
Semifinals	CF378-D2-B	—
Towers	CF479-D2-B	—

Problem	Problem Link	Video / Solution
Gerald is into Art	CF560-D2-B	—

■ **WATCH: Watch - Algebra - Number Bases and Polynomials**

Problem	Problem Link	Video / Solution
To Carry or not to Carry	UVA 10469	—
Beat the Spread!	UVA 10812	—
498-bis	UVA 10268	—
Summation of Polynomials	UVA 10302	—
Polly the Polynomial	UVA 498	—

■ **WATCH: Watch - Algebra - Patterns in Sequences**

Problem	Problem Link	Video / Solution
Odd Sum	UVA 10783	—
R U Kidding Mr. Feynman?	UVA 10509	—
Cleaning Robot	SPOJ CLEANRBT	—
Wandering Queen	SPOJ QUEEN	Sol to read
Spreadsheet	UVA 196	—
—	HACKR sherlock-and-probability	—

■ **WATCH: Watch - Algebra - Summations**

■ **WATCH: Watch - Algebra - Basic Matrix Operations**

Problem	Problem Link	Video / Solution
Little Girl and Maximum Sum	CF276-D2-C	—
Searching for Graph	CF402-D2-C	—
Sereja and Algorithm	CF368-D2-C	—
Replacement	CF570-D2-C	—
Four Segments	CF14-D2-C	—
Flying Saucer Segments	CF227-D2-C	—
Vasya and Petya's Game	CF577-D2-C	—
XOR and OR	CF282-D2-C	—
Round Table Knights	CF71-D2-C	—
Kefa and Park	CF580-D2-C	Video Solution - Solver to be (Java)
Progress Bar	CF71-D2-B	—
Special Offer! Super Price 999 Bourles!	CF219-D2-B	—
Jury Size	CF254-D2-B	—

■ **WATCH: Watch - Thinking - Problem Abstraction**

■ **WATCH: Watch - Thinking - Problem Reverse**

■ **WATCH: Watch - Search Techniques - Backtracking**

Problem	Problem Link	Video / Solution
Graph Coloring	UVA 193	Video Solution - Eng Mostafa Saad
23 out of 5	UVA 10344	Video Solution - Eng Mohamed Nasser

Problem	Problem Link	Video / Solution
8 Queens Chess Problem	UVA 750	Video Solution - Eng Ayman Salah
Assemble	UVA 12124	—
—	UVA 10056	Sol to read
Dishonest Sellers	CF779-D2-C	Video Solution - Solver to be (Java)
Magic Formulas	CF424-D2-C	—
Pythagorean Triples	CF707-D2-C	—
Gerald's Hexagon	CF560-D2-C	—
Points on Line	CF252-D2-C	—
Find Maximum	CF353-D2-C	—
Secrets	CF334-D2-C	—
Milking cows	CF384-D2-C	—
Primes on Interval	CF237-D2-C	—
Cardiogram	CF435-D2-C	—
Jzzhu and Sequences	CF450-D2-B	—
Simple Game	CF570-D2-B	—
Prime Matrix	CF271-D2-B	—

■ **WATCH: Watch - DP - Subset Style**

Problem	Problem Link	Video / Solution
Vacation	UVA 10192	Explained in the tutorial videos
Strategic Defense Initiative	UVA 497	Explained in the tutorial videos
Dividing coins	UVA 562	Video Solution - Eng Ayman Salah

■ **WATCH: Watch - DP - Consecutive Ranges Style**

Problem	Problem Link	Video / Solution
The Blocks Problem	UVA 101	—
Divisibility	UVA 10036	—
Perfect Hash	UVA 188	—
—	—	Sol to read
—	UVa 11181	Sol to read
Bombs	CF350-D2-C	—
Rational Resistance	CF344-D2-C	—
Appleman and Toastman	CF462-D2-C	Video Solution - Solver to be (Java)
Dima and Staircase	CF272-D2-C	—
k-Multiple Free Set	CF275-D2-C	—
Little Pony and Expected Maximum	CF454-D2-C	—
Boredom	CF456-D2-C	—
Wizards and Trolleybuses	CF168-D2-C	—
Anya and Smartphone	CF518-D2-C	—
Polycarpus' Dice	CF534-D2-C	—
Print Check	CF631-D2-B	—
Playing Cubes	CF257-D2-B	—

Problem	Problem Link	Video / Solution
T-primes	CF230-D2-B	—

■ **WATCH:** Watch - DP - Nested Ranges Style

■ **WATCH:** Watch - DP - General Ranges Style

Problem	Problem Link	Video / Solution
String to Palindrome	UVA 10739	Explained in the tutorial videos
Creating Palindrome	UVA 11753	Video Solution - Eng Aya Elymany
Again Palindrome	UVA 10617	Sol to read
Exploring Pyramids	UVA 1362	Video Solution - Eng Ayman Salah
Cutting Sticks	UVA 10003	—
Trouble of 13-Dots	UVA 10819	—
Optimal Array Multiplication Sequence	UVA 348	Sol to read
Accordian Patience	UVA 127	Video Solution - Eng Moaz Rashad
Software CRC	UVA 128	Video Solution - Eng Moaz Rashad
Maze	CF378-D2-C	—
Polycarp at the Radio	CF723-D2-C	—
Thor	CF705-D2-C	—
Hard problem	CF706-D2-C	—
Unusual Product	CF405-D2-C	—
Palindrome Transformation	CF486-D2-C	—
Chain Reaction	CF608-D2-C	—
Removing Columns	CF496-D2-C	Video Solution - Eng Mostafa Saad
Newspaper Headline	CF92-D2-C	—
Crazy Town	CF499-D2-C	Video Solution - Eng Mostafa Saad
Queue	CF490-D2-B	—
Vika and Squares	CF610-D2-B	—
Cosmic Tables	CF222-D2-B	—

■ **WATCH:** Watch - Thinking - Incrementally

■ **WATCH:** Watch - Thinking - Problem Domain re-interpretation

■ **WATCH:** Watch - Number Theory - Factorization

Problem	Problem Link	Video / Solution
Prime Factors	UVA 583	—
Count the factors	UVA 10699	—
Perfection	UVA 382	—
Mr. Azad and his Son	UVA 10490	Sol to read
Perfect P-th Powers	UVA 10622	Video Solution - Eng Moaz Rashad
Code Refactoring	UVA 10879	—
Prime Land	UVA 516	—
Text Editor	CF253-D2-C	—
Alternative Thinking	CF604-D2-C	—

Problem	Problem Link	Video / Solution
Tennis Championship	CF735-D2-C	—
Shifts	CF230-D2-C	—
Guess Your Way Out!	CF507-D2-C	Video Solution - Eng Mostafa Saad
Frames	CF94-D2-C	—
Alyona and the Tree	CF682-D2-C	—
Biathlon	CF84-D2-C	—
Marina and Vasya	CF584-D2-C	—
The Two Routes	CF602-D2-C	Video Solution - Solver to be (Java)
Divide by Three	CF792-D2-C	Video Solution - Solver to be (Java)
Little Pigs and Wolves	CF116-D2-B	—
Covered Path	CF534-D2-B	—
Facetook Priority Wall	CF75-D2-B	—
Perfect Cubes	UVA 386	—
How Many Points of ?	UVA 10790	—
Nature	UVA 10685	—
Factovisors	UVA 10139	Sol to read
Changing Maze	SPOJ CHMAZE	Sol to read
Fractions Again?!	UVA 10976	Sol to read
Philosophers Stone	SPOJ BYTESM2	Video Solution - Solver to be (Java)
Treats for the Cows	SPOJ TRT	Video Solution - Solver to be (Java)
Cut Ribbon	CF189-D2-A	Video Solution - Solver to be (Java)

■ **WATCH: [Watch - Probability - First 9 videos](#)**

Problem	Problem Link	Video / Solution
—	—	Sol to read
Cows and Cars	UVA 10491	Sol to read
What is the Probability?	UVA 10056	Sol to read
Let's Dance	UVA 10218	Sol to read
Probability Given	UVA 11181	Sol to read
Another lottery	UVA 11628	Sol to read
Airplane	UVA 12461	Sol to read
Hacker, pack your bags!	CF822-D2-C	Video Solution - Solver to be (Java)
The Meaningless Game	CF834-D2-C	Video Solution - Solver to be (Java)
Star sky	CF835-D2-C	Video Solution - Solver to be (Java)
Two TVs	CF845-D2-C	Video Solution - Solver to be (Java)
Diverse Permutation	CF483-D2-C	—
Replacement	CF136-D2-C	—
Homework	CF102-D2-C	—
Little Elephant and Problem	CF221-D2-C	—
Developing Skills	CF581-D2-C	—
Maxim and Discounts	CF262-D2-C	—

Problem	Problem Link	Video / Solution
Fox and Box Accumulation	CF389-D2-C	—
Ice Skating	CF218-D2-C	—
Valera and Tubes	CF441-D2-C	—
Secret	CF271-D2-C	—
They Are Everywhere	CF701-D2-C	—
Monitor	CF16-D2-C	—
System Administrator	CF22-D2-C	—
Lucky Sum	CF122-D2-C	—
NP-Hard Problem	CF688-D2-C	—
Vladik and fractions	CF743-D2-C	—
Case of Matryoshkas	CF556-D2-C	—
Vanya and Label	CF677-D2-C	—
Exams	CF479-D2-C	—
Purification	CF330-D2-C	—
Division into Teams	CF149-D2-C	—
Disposition	CF49-D2-C	—
Mashmokh and Numbers	CF415-D2-C	—
Statues	CF129-D2-C	—
Inna and Huge Candy Matrix	CF400-D2-C	—
Anagram Search	CF144-D2-C	—
Greg and Array	CF296-D2-C	—
Ilya and Sticks	CF525-D2-C	—
Day at the Beach	CF599-D2-C	—
The Child and Toy	CF437-D2-C	—
Perfect Pair	CF318-D2-C	—
Another Problem on Strings	CF165-D2-C	—
Socks	CF731-D2-C	—
Valera and Elections	CF369-D2-C	—

Sheet C2 — Codeforces Div2-C (Part 2)

■ **WATCH:** Watch - Thinking - Search Space and Output Analysis

■ **WATCH:** Watch - Thinking - Observations Discovery

■ **WATCH:** Watch - Game Theory - Intro

Problem	Problem Link	Video / Solution
Win or Freeze	CF151-D2-C	Video Solution - Eng Mostafa Saad
Puzzle	UVA 227	—
Pyramids	SPOJ PIR	Sol to read
Power of Cryptography	UVA 113	Sol to read
—	CF105-D12-B	—
Modified GCD	CF75-D2-C	Video Solution - Eng Mostafa Saad
Alyona and mex	CF740-D2-C	Video Solution - Eng Mostafa Saad
Color Stripe	CF219-D2-C	—
Hamburgers	CF371-D2-C	—
Wet Shark and Flowers	CF621-D2-C	—
Predict Outcome of the Game	CF451-D2-C	—
MUH and House of Cards	CF471-D2-C	—
Heap Operations	CF681-D2-C	—
Balls and Boxes	CF260-D2-C	Video Solution - Eng Mostafa Saad
Alice and Bob	CF347-D2-C	Video Solution - Eng Mohamed Nasser
Mahmoud and Ehab and the wrong algorithm	CF959-D2-C	Video Solution - Eng Mohamed Salah
Balls Game	CF430-D2-B	—
Permutations	CF124-D2-B	—
Magical Array	CF84-D2-B	—

■ **WATCH:** Watch - Thinking - Misc - Solution Verification - Implementation

■ **WATCH:** Watch - Graph Theory - Dijkstra

Problem	Problem Link	Video / Solution
Jugs	UVA 571	Video Solution - Eng Mostafa Saad
Sending email	UVA 10986	—
Lift Hopping	UVA 10801	—
Shopping	SPOJ SHOP	—
Deciding victory in Go	UVA 852	—
Divisors	UVA 294	—
The Cats and the Mouse	SPOJ CATM	Sol
Ordering	UVA 872	—
Bulls and Cows	CF63-D2-C	—
George and Number	CF387-D2-C	—
Xor-tree	CF430-D2-C	—
Median Smoothing	CF591-D2-C	—

Problem	Problem Link	Video / Solution
Coloring Trees	CF711-D2-C	—
Robbers' watch	CF686-D2-C	—
Clear Symmetry	CF202-D2-C	—
Malek Dance Club	CF320-D2-C	—
Sereja and Swaps	CF426-D2-C	—
Trees	CF58-D2-C	—
Sereja and Mirroring	CF426-D2-B	—
Restoring Painting	CF675-D2-B	—
Amr and The Large Array	CF558-D2-B	—

■ **WATCH: Watch - Computational Geometry - Lines Intersections**

Problem	Problem Link	Video / Solution
Gleaming the Cubes	UVA 737	Sol to read
Intersecting Line Segments	UVA 866	—

■ **WATCH: Watch - Computational Geometry - Circles**

Problem	Problem Link	Video / Solution
The Circumference of the Circle	UVA 438	—
Points in Figures: Rectangles and Circles	UVA 477	—
Square Pegs And Round Holes	UVA 356	Sol to read
Divisibility of Factors	UVA 10484	Sol to read
Prime Permutation	CF124-D2-C	—
Hometask	CF155-D2-C	—
Arithmetic Progression	CF382-D2-C	—
Terse princess	CF148-D2-C	Video Solution - Eng Mohamed Nasser
Help Farmer	CF143-D2-C	—
Hacking Cypher	CF490-D2-C	—
Dreamoon and Sums	CF476-D2-C	Video Solution - Eng Mostafa Saad
Try and Catch	CF195-D2-C	—
Primes or Palindromes?	CF569-D2-C	—
View Angle	CF257-D2-C	—
Flag Day	CF357-D2-B	—
Little Pony and Sort by Shift	CF454-D2-B	—
Two Tables	CF228-D2-B	—

■ **WATCH: Watch - Thinking - Error Inspection - History - Contest Strategy**

■ **WATCH: Watch - DP - Building Output**

Problem	Problem Link	Video / Solution
Unidirectional TSP	UVA 116	—
Make Palindrome	UVA 10453	—
Fast Food	UVA 662	—
Palindromic Subsequence	UVA 11404	—

Problem	Problem Link	Video / Solution
Gone Fishing	UVA 757	Sol to read
Special Olympics	CF199-D2-B	—
Rings and Glue	UVA 10301	Sol to read
—	CF754-D2-C	Video Solution - Solver to be (Java)

■ **WATCH: [Watch - DP - Counting](#)**

Problem	Problem Link	Video / Solution
k-Tree	CF431-D2-C	Video Solution - Solver to be (Java)
Caesar's Legions	CF118-D2-D	—
UnsealTheSafe	—	—
DiceGames	—	—
StringWeightDiv2	—	—
No to Palindromes!	CF465-D2-C	—
Mike and Chocolate Thieves	CF689-D2-C	—
Triangle	CF408-D2-C	—
To Add or Not to Add	CF231-D2-C	—
Number of Ways	CF466-D2-C	Video Solution - Solver to be (Java)
Queue	CF141-D2-C	—
Grammar Lessons	CF114-D2-C	—
Magical Boxes	CF270-D2-C	—
Knight Tournament	CF357-D2-C	—
Find Pair	CF160-D2-C	—
Appleman and Card Game	CF462-D2-B	—
Multitasking	CF384-D2-B	—
Non-square Equation	CF233-D2-B	—

■ **WATCH: [Watch - Thinking - Let's Put All Together](#)**

■ **WATCH: [Watch - DP - Table Method](#)**

■ **WATCH: [Watch - Graph Theory - Floyd Warshal](#)**

Problem	Problem Link	Video / Solution
Non-Stop Travel	UVA 341	—
Frogger	UVA 534	Sol to read
Identifying Concurrent	UVA 334	—
Numbering Paths	UVA 125	Sol
Jack Straws	UVA 273	Sol to read
Longest Match	UVA 10100	—
Isolated Segments	UVA 11343	Sol to read
Highways	SPOJ HIGHWAYS	—
Counting	UVA 10198	—
Mafia	CF349-D2-C	—
Sereja and Prefixes	CF381-D2-C	—

Problem	Problem Link	Video / Solution
About Bacteria	CF199-D2-C	—
DNA Alignment	CF520-D2-C	—
Vasily the Bear and Sequence	CF336-D2-C	—
Geometric Progression	CF567-D2-C	—
Mittens	CF370-D2-C	—
Prime Swaps	CF432-D2-C	—
Watering Flowers	CF617-D2-C	—
Quiz	CF337-D2-C	—
Secret Combination	CF496-D2-B	—
MUH and Important Things	CF471-D2-B	—
Lucky Mask	CF146-D2-B	—

■ **WATCH:** [Watch - Measuring Algorithms Performance - 2](#)

■ **WATCH:** [Watch - Graph Theory - Tree Diameter and Isomorphism](#)

Problem	Problem Link	Video / Solution
PT07Z	SPOJ PT07Z	—
Roads in the North	UVA 10308	—
Subway tree systems	LIVEARCHIVE 2935	Sol to read
Bracket Sequence	CF224-D2-C	—
Recycling Bottles	CF672-D2-C	—
Fancy Number	CF118-D2-C	—
Matrix	CF365-D2-C	—
Glass Carving	CF527-D2-C	—
Shaass and Lights	CF294-D2-C	Video Solution - Eng Mostafa Saad
Journey	CF721-D2-C	—
Captain Marmot	CF474-D2-C	Video Solution - Eng Mostafa Saad
The Big Race	CF592-D2-C	—
Molly's Chemicals	CF776-D2-C	Video Solution - Solver to be (Java)
Anatoly and Cockroaches	CF719-D2-B	—
Kolya and Tandem Repeat	CF443-D2-B	—
Opposites Attract	CF131-D2-B	—
Railway	UVA 10263	Sol to read
Factorial Factors	UVA 884	—
Intersection	UVA 191	Sol to read
Wifi Access	UVA 12748	Sol to read
Lining Up	UVA 270	Video Solution - Eng Mohamed Nasser. Don't Code $O(N^3)$
Pouring water	SPOJ POUR1	—

■ **WATCH:** [Watch Video - Expected Value](#)

Problem	Problem Link	Video / Solution
God, Save me	UVA 10777	Sol to read

Problem	Problem Link	Video / Solution
—	CF839-D2-C	—
—	CF454-D2-C	—
—	HACKR lazy-sorting	—
Checkposts	CF427-D2-C	—
Literature Lesson	CF139-D2-C	—
Arpa's loud Owf and Mehrdad's evil plan	CF742-D2-C	—
Parity Game	CF298-D2-C	—
Beauty Pageant	CF246-D2-C	—
Heroes	CF80-D2-C	—
Dynasty Puzzles	CF192-D2-C	—
Buns	CF106-D2-C	—
Counting Kangaroos is Fun	CF373-D2-C	—
Corporation Mail	CF56-D2-C	—
Little Elephant and Interval	CF205-D2-C	—
Sereja and Contest	CF315-D2-C	—
Vasya and Robot	CF355-D2-C	—
Hockey	CF96-D2-C	—
Petya and File System	CF66-D2-C	—
Kyoya and Colored Balls	CF554-D2-C	—
George and Job	CF467-D2-C	—
Harmony Analysis	CF610-D2-C	—
Anton and Making Potions	CF734-D2-C	—
Table Decorations	CF478-D2-C	—
Message	CF157-D2-C	—
Wilbur and Points	CF596-D2-C	—
Cows and Sequence	CF284-D2-C	—
Ladder	CF279-D2-C	—
Not Wool Sequences	CF239-D2-C	—
Anagram	CF254-D2-C	—
DZY Loves Sequences	CF447-D2-C	—
DZY Loves Physics	CF445-D2-C	—
Misha and Forest	CF501-D2-C	—
Jzzhu and Chocolate	CF450-D2-C	—
Report	CF631-D2-C	—
Bear and Prime Numbers	CF385-D2-C	—
Robbery	CF90-D2-C	—
Vasya and Basketball	CF493-D2-C	—
Vanya and Scales	CF552-D2-C	—
Pashmak and Buses	CF459-D2-C	—

Sheet D1 — Codeforces Div2-D (Part 1)

Problem	Problem Link	Video / Solution
Solitaire	CF71-D2-D	—
Vasya and Types	CF88-D2-D	—
Petya and Divisors	CF112-D2-D	—
Dividing Island	CF63-D2-D	—
Long Jumps	CF479-D2-D	—
Vasya and Chess	CF493-D2-D	—
Flowers	CF474-D2-D	Video Solution - Solver to be (Java)
Queue	CF545-D2-D	Video Solution - Solver to be (Java)
Vanya and Triangles	CF552-D2-D	—
Dima and Bacteria	CF400-D2-D	—
Tourist Problem	CF340-D2-C	—
Lorenzo Von Matterhorn	CF697-D2-C	—
Restore Graph	CF404-D2-C	—

■ WATCH: [Watch - Data Structures - Segment Tree \(2 vid\)](#)

Problem	Problem Link	Video / Solution
Interval Product	UVA 12532	—
Potentiometers	LIVEARCHIVE 2191	—
Halt The War	SPOJ CDC12_H	—
Counting Primes	SPOJ CNTPRIME	—
Light Switching	SPOJ LITE	—
Horrible Queries	SPOJ HORRIBLE	—
RMQ with Shifts	UVA 12299	See sscanf and sprintf usage
R2D2 and Droid Army	CF514-D2-D	Solve using rmq
Ahoy, Pirates!	UVA 11402	—
Circular RMQ	CF52-D12-C	—
Brackets	SPOJ BRCKTS	—
Present	CF460-D2-C	—
A Famous City	SPOJ CITY2	Sol
MessageMess	—	—
DiceGames	—	—
Mirror, Mirror	UVA 466	—
Pick up sticks	UVA 11686	—
Maximum Sum	SPOJ KGSS	—
Soldier and Number Game	CF546-D2-D	—
One-Dimensional Battle Ships	CF567-D2-D	Video Solution - Solver to be (Java)
Vanya and Computer Game	CF492-D2-D	—
Lawnmower	CF116-D2-D	—
Treasure Island	CF106-D2-D	—

Problem	Problem Link	Video / Solution
Quantity of Strings	CF151-D2-D	—
Eternal Victory	CF61-D2-D	—
Jumping on Walls	CF199-D2-D	—
Alyona and Strings	CF682-D2-D	—
Array Division	CF808-D2-D	Video Solution - Solver to be (Java)
Hiring Staff	CF216-D2-C	—
Tavas and Karafs	CF535-D2-C	—
Permutations	CF189-D2-C	Sol

■ **WATCH: *Watch - Two pointers technique***

Problem	Problem Link	Video / Solution
Spider's Web	CF216-D2-D	—
Chips	CF334-D2-D	—
Vasya and String	CF676-D2-C	—
We Ship Cheap	UVA 762	—
Symmetry	UVA 1595	—
The SetStack Computer	LiveArchive 3634	—
Database	UVA 1592	—
Can you answer these queries I	SPOJ GSS1	—
Can you answer these queries III	SPOJ GSS3	—
Common Divisors	CF182-D2-D	Video Solution - Solver to be (Java)
Mahmoud and a Dictionary	CF766-D2-D	Video Solution - Solver to be (Java)
An overnight dance in discotheque	CF814-D2-D	Video Solution - Solver to be (Java)
Polyline	CF617-D2-D	—
Queue	CF92-D2-D	—
Tree Construction	CF675-D2-D	—
Exams	CF732-D2-D	—
Phillip and Trains	CF586-D2-D	—
Palindromes	CF137-D2-D	—
Sum	CF257-D2-D	—
Plant	CF186-D2-C	—
Reberland Linguistics	CF667-D2-C	—
Lucky Permutation	CF287-D2-C	—
—	—	DP - Probability
Dice Throwing	UVA 10759	Sol
TestBettingStrategy	—	—
Hopeless Coach	TJU 3051	—
Collecting Bugs	PKU 2096	Sol
Andrey and Problem	CF443-D2-D	—
France '98	UVA 542	Sol
Tribbles	UVA 11021	Sol

Problem	Problem Link	Video / Solution
Hats	UVA 12024	Sol
Tennis contest	UVA 12457	Sol
Water Falls	UVA 833	Sol
Number Sequence	UVA 10706	—
Is It A Tree?	UVA 615	—
Help R2-D2!	SPOJ HELPR2D2	—
Calendar	CF58-D2-D	—
Colliders	CF155-D2-D	—
Suspects	CF157-D2-D	—
Lakes in Berland	CF723-D2-D	—
Three Logos	CF581-D2-D	—
Programming Language	CF200-D2-D	—
Journey	CF43-D2-D	—
Good Sequences	CF265-D2-D	—
Handshakes	CF534-D2-D	—
Working out	CF430-D2-D	—
Party	CF116-D2-C	—
Cupboard and Balloons	CF342-D2-C	—
Cycles	CF233-D2-C	—
—	—	DP - Masks (2 vid)
Pebble Solitaire	UVA 10651	—
Kefa and Dishes	CF580-D2-D	—
Permutations	SPOJ PERMUT1	—
Assignments	SPOJ ASSIGN	—
Zeros and Ones	UVA 12063	—
Islands and Bridges	TJU 1189	—
Count the Faces.	UVA 10178	—
Is There A Second Way Left?	UVA 10462	—
Equation	UVA 727	—
Safe	CF47-D2-D	—
Beverages	UVA 11060	—
Robot Rapping Results Report time	CF645-D12-D	—
Directed Roads	CF711-D2-D	—
Block Tower	CF327-D2-D	—
Pawn	CF41-D2-D	—
Gargari and Permutations	CF463-D2-D	—
A and B and Interesting Substrings	CF519-D2-D	—
As Fast As Possible	CF701-D2-D	—
Chloe and pleasant prizes	CF743-D2-D	—
Doctor	CF84-D2-D	—

Problem	Problem Link	Video / Solution
Maximum Submatrix 2	CF376-D2-D	Video Solution - Solver to be (Java)
Help General	CF143-D2-D	—
Roads in Berland	CF25-D2-C	—
Photographer	CF203-D2-C	—
LCM Challenge	CF236-D2-C	—
—	—	String Processing - Trie
Search in the dictionary!	SPOJ DICT	—
Disk Tree	UVA 1556	—
Phone List	SPOJ PHONELST	—
Cellphone Typing	UVA 12526	—
Vasiliy's Multiset	CF706-D2-D	—
Exchange Rates	UVA 10113	—
Wavio Sequence	UVA 10534	—
LCM Cardinality	UVA 10892	—
Central Post Office	UVA 12379	—
Permalex	UVA 153	—
Planets	CF230-D2-D	—
Missile Silos	CF144-D2-D	—
Volleyball	CF96-D2-D	—
Lazy Student	CF606-D2-D	—
Multiplication Table	CF448-D2-D	Video Solution - Solve to be (Java)
Divisible by Seven	CF376-D2-C	—
Devu and Partitioning of the Array	CF439-D2-C	—
Arthur and Table	CF557-D2-C	—

Sheet D2 — Codeforces Div2-D (Part 2)

Problem	Problem Link	Video / Solution
—	—	String Processing - KMP (2 vid)
Oulipo	PKU 3461	—
A Needle in the Haystack	SPOJ NHAY	—
Finding the Tesseract	SPOJ TESSER	—
Period	SPOJ PERIOD	—
Prefixes and Suffixes	CF432-D2-D	—
Tavas and Malekas	CF535-D2-D	—
Password	CF127-D2-D	—
Another Game With Numbers	SPOJ NGM2	—
Word Transformation	UVA 429	—
The PATH	UVA 601	Video Solution - Eng Mostafa Saad
Be Efficient	UVA 11155	—
Vertex Cover	SPOJ PT07X	—
First Digit Law	CF54-D12-C	—
—	CF500-D12-D	—
—	HACKR vertical-sticks	—
Long Path	CF408-D2-D	—
Robin Hood	CF672-D2-D	—
Toy Sum	CF405-D2-D	—
Dreamoon and Sets	CF476-D2-D	—
End of Exams	CF94-D2-D	—
Equivalent Strings	CF560-D2-D	Sol - intended
Count Good Substrings	CF451-D2-D	—
Mushroom Scientists	CF186-D2-D	—
Analyzing Polyline	CF195-D2-D	—
Mashmokh and ACM	CF415-D2-D	—
Bear and Prime 100	CF680-D2-C	—
24 Game	CF469-D2-C	—
Team	CF401-D2-C	—
—	—	DP - Games (2 vid)
Bachet's Game	UVA 10404	—
EllysCheckers	—	—
RowAndCoins	—	—
BagsOfGold	—	—
Bag of mice	CF148-D2-D	—
Laser Phones	SPOJ MLASERP	—
MELE3	SPOJ MELE3	—
Roads	SPOJ ROADS	—

Problem	Problem Link	Video / Solution
The Tree Root	UVA 10459	—
SKYLINE	UVA 1232	Sol
Ordering the Soldiers	SPOJ ORDERS	Sol
Playlist	CF268-D2-E	Sol
The Dueling Philosophers Problem	LIVEARCHIVE 6195	—
Unbearable Controversy of Being	CF489-D2-D	—
Fix a Tree	CF699-D2-D	—
Roads not only in Berland	CF25-D2-D	—
Dima and Lisa	CF584-D2-D	—
Little Girl and Maximum XOR	CF276-D2-D	See editorials
Two Strings	CF224-D2-D	—
Queue	CF353-D2-D	—
Maxim and Array	CF721-D2-D	—
Big Maximum Sum	CF75-D2-D	—
Ants	CF318-D2-D	—
Efim and Strange Grade	CF719-D2-C	—
Football Championship	CF200-D2-C	—
Given Length and Sum of Digits...	CF489-D2-C	—
Trip Routing	UVA 186	—
Scheduling Lectures	UVA 607	Sol
Weird Function	SPOJ WEIRDFN	Sol
The ? 1 ? 2 ? ... ? n = k problem	UVA 10025	—
Dictionary Subsequences	SPOJ DICTSUB	—
GONDOR	SPOJ GONDOR	—
Interesting Array	CF483-D2-D	—
Minimum Transport Cost	UVA 523	Sol
Jimmi's Riddles	UVA 10058	Sol
Friends and Subsequences	CF689-D2-D	—
Sum of Squares with Segment Tree	SPOJ SEGSQLRSS	Sol
Travel in Desert	UVA 10816	—
Almost Union-Find	—	—
—	CF513-D12-C	Sol
Cow Program	CF284-D2-D	—
Random Task	CF431-D2-D	—
Biridian Forest	CF330-D2-D	—
Dot	CF69-D2-D	—
Greg and Graph	CF296-D2-D	—
Russian Roulette	CF104-D2-D	—
Vessels	CF371-D2-D	Video Solution - Solver to be (Java)
Bicycle Race	CF659-D2-D	—

Problem	Problem Link	Video / Solution
Greenhouse Effect	CF270-D2-D	—
Rectangle and Square	CF136-D2-D	—
Pocket Book	CF152-D2-C	—
Levko and Array Recovery	CF361-D2-C	—
Ice Cave	CF540-D2-C	—
Robbery	UVA 707	—
The Errant Physicist	UVA 126	Sol
ICPC Scoreboard	SPOJ ICPCS	—
Brackets sequence	UVA 1626	Sol
Unique World	UVA 10448	Video Solution - Eng Mostafa Saad
Bad Luck Island	CF540-D2-D	—
Game of Euler	—	Sol
Shopping Trip	—	Sol
Hotel booking	—	—
Shopping Trip	—	Sol
Hit Ball	CF203-D2-D	—
Sereja and Anagrams	CF368-D2-D	Sol
Choosing Capital for Treeland	CF219-D2-D	—
Coloring Brackets	CF149-D2-D	Sol
Changing a String	CF56-D2-D	—
Dima and Hares	CF358-D2-D	—
Two Paths	CF14-D2-D	—
Hot Days	CF215-D2-D	—
Cycle in Graph	CF263-D2-D	—
Fixing Typos	CF363-D2-C	—
Cutting Figure	CF194-D2-C	—
Escape from Stones	CF265-D2-C	—
—	—	Geometry - Simple and Convex Polygons
—	—	Geometry - Polygon Area - Centroid - Cut
BestTriangulation	—	—
Xrange's Pancakes	HACKR xrange-and-pizza	Sol
Trees on My Island	UVA 10088	—
Packing polygons	UVA 10005	Sol
Video Surveillance	UVA 588	See polygon cut solution
The Most Potent Corner	UVA 10264	—
—	CF100200-GYM-C	—
—	—	Geometry - Point in polygon
—	—	Sol
—	—	Sol
—	—	Sol

Problem	Problem Link	Video / Solution
Polygons	UVA 137	Sol
—	—	Graph Theory - Maximum Flow (2 vid)
A Plug for UNIX	UVA 753	—
Intergalactic Map	SPOJ IM	Sol
Potholders	SPOJ POTHOLE	—
Dining	TJU 2823	Sol
Power Transmission	UVA 10330	—
Gopher II	UVA 10080	—
Software Allocation	UVA 259	—
Fence Obstacle Course	PKU 2374	—
River Crossing	—	Sol
—	—	Sol
—	—	Sol

Sheet D3 — Codeforces Div2-D (Part 3)

Problem	Problem Link	Video / Solution
—	—	Graph Theory - SCC (2 vid)
The Bottom of a Graph	SPOJ BOTTOM	—
Perfect Election	LIVEARCHIVE 4185	—
Road Networks	LIVEARCHIVE 4262	—
Checkposts	CF427-D2-C	—
Test	UVA 10731	—
Theseus and labyrinth	CF676-D2-D	—
Cunning Gena	CF418-D1-B	—
Sabotage	UVA 10480	—
Gifts by the List	CF681-D2-D	—
Olympiad	CF222-D2-D	—
DZY Loves Modification	CF447-D2-D	Think about a proof
Mike and Feet	CF548-D2-D	—
Permutation Sum	CF285-D2-D	—
Arpa's weak amphitheater and Mehrdad's valuable Hoses	CF742-D2-D	—
Special Grid	CF435-D2-D	—
Roman and Numbers	CF401-D2-D	—
Persistent Bookcase	CF707-D2-D	Sol
Regular Bridge	CF550-D2-D	—
Almost Arithmetical Progression	CF255-D2-C	—
Title	CF59-D2-C	—
Treasure	CF495-D2-C	—
End of Fun	SPOJ DCEPC12E	—
Grammar Evaluation	UVA 622	Sol
Find the Winning Move	UVA 10111	Sol
Check the difficulty of problems	PKU 2151	Sol
Proving Equivalences	UVA 12167	—
DDF	UVA 547	—
Dominos	UVA 11504	Sol
Winning Streak	UVA 11176	—
Red-Green Towers	CF478-D2-D	—
Renting Bikes	CF363-D2-D	—
Lucky Number 2	CF146-D2-D	—
Digits Permutations	CF139-D2-D	—
Tennis Game	CF496-D2-D	—
Bubble Sort Graph	CF340-D2-D	—
Polo the Penguin and Houses	CF289-D2-D	—
Upgrading Array	CF402-D2-D	—

Problem	Problem Link	Video / Solution
Demonstration	CF192-D2-D	—
Naughty Stone Piles	CF227-D2-D	—
Game	CF69-D2-C	—
Ciel and Robot	CF322-D2-C	—
Plus and Square Root	CF716-D2-C	—
Boxes in a Line	UVA 12657	Sol
Antenna Placement	UVA 10349	Sol
Multifactorials	UVA 11347	—
Crimewave	UVA 563	—
Jzzhu and Cities	CF450-D2-D	—
Primitive Root	SPOJ PROOT	—
Of Zorcs and Axes	CF101149-GYM-G	—
Connected Components	CF292-D12-D	—
AND Rounds	SPOJ ANDROUND	—
Campus Roads	UVA 11473	Sol
The Child and Zoo	CF437-D2-D	—
Image Preview	CF651-D2-D	—
Maximum Xor Secondary	CF281-D2-D	—
Psychos in a Line	CF320-D2-D	—
Ilya and Roads	CF313-D2-D	—
Mr. Bender and Square	CF255-D2-D	—
Minesweeper 1D	CF404-D2-D	—
Fish Weight	CF298-D2-D	—
Guess That Car!	CF202-D2-D	—
T-decomposition	CF237-D2-D	—
Wizards and Huge Prize	CF168-D2-D	—
Dima and Salad	CF366-D2-C	—
Gennady the Dentist	CF586-D2-C	—
Cinema	CF670-D2-C	—
Arbitrage	UVA 104	Sol
Random Task	CF431-D2-D	—
Black Box	UVA 501	Use STL. Sol
Expressions	UVA 11234	Sol
Showstopper	SPOJ MSE07E	Read SPOJ users' comments about IO
Tobo or not Tobo	SPOJ ANARC08A	Sol
Sum-up the Primes	UVA 10419	—
Largest Rectangle in a Histogram	SPOJ HISTOGRAM	Sol. Don't implement as adhoc/greedy/Pure STL. Use a data structure.
Factors and Multiples	UVA 11159	—
KingdomReorganization	—	—

Problem	Problem Link	Video / Solution
The Problem with the Problem Setter	UVA 10092	—
Psycho	SPOJ PSYCHON	—
Minimal Ratio Tree	LIVEARCHIVE 4326	—
RACING	UVA 1234	Sol
ActivateGame	—	—
Pair of Numbers	CF359-D2-D	—
Nuts for nuts	—	—
End up with More Teams	—	—
Probability	UVa 11346	Sol
AlgoRace	CF189-D2-D	Sol
Moodular Arithmetic	CF604-D2-D	Sol
Anton and Chess	CF734-D2-D	—
Lucky Transformation	CF122-D2-D	—
Mysterious Present	CF4-D2-D	—
Alternating Current	CF344-D2-D	—
Boring Partition	CF239-D2-D	—
Spongebob and Squares	CF599-D2-D	—
How many trees?	CF9-D2-D	—
Kostya the Sculptor	CF733-D2-D	—
Gargari and Bishops	CF463-D2-C	—
Cthulhu	CF104-D2-C	—
Anya and Ghosts	CF508-D2-C	—
Square Subsets	CF448-D2-C	—
Angry Programmer	UVA 11506	—
The New Rule in Euphonia	UVA 10742	—
Fast Matrix Operations	UVA 11992	—
Multiples of 3	SPOJ MULTQ3	—
TimeTravellingSalesman	—	—
March of the Penguins	UVA 12125	Sol
WTommy's Trouble	TJU 2233	—
PeopleYouMayKnow	—	Don't use DP. Sol
The Game of 31	UVA 10578	—
Can you answer these queries IV	SPOJ GSS4	—
Area	TJU 1011	Sol
Volatile Kite	CF801-D2-D	Video Solution - Eng Youssef El Ghareeb
Antifloyd	UVA 10987	Sol
Messenger	CF631-D2-D	—
—	CF280-D1-C	—
—	CF110-D2-D	—
—	CF163-D12-C	—

Problem	Problem Link	Video / Solution
Infinite Maze	CF197-D2-D	—
Jeff and Furik	CF352-D2-D	—
Colorful Graph	CF246-D2-D	—
Sagheer and Kindergarten	CF812-D2-D	Video Solution - Solver to be (Java)
Dispute	CF242-D2-D	—
Remainders Game	CF688-D2-D	—
String Mark	CF448-D2-D	—